

FORMAL QUERY LANGUAGES

RELATIONAL ALGEBRA

TUPLE RELATIONAL
CALCULUS

DOMAIN RELATIONAL
CALCULUS

Agenda

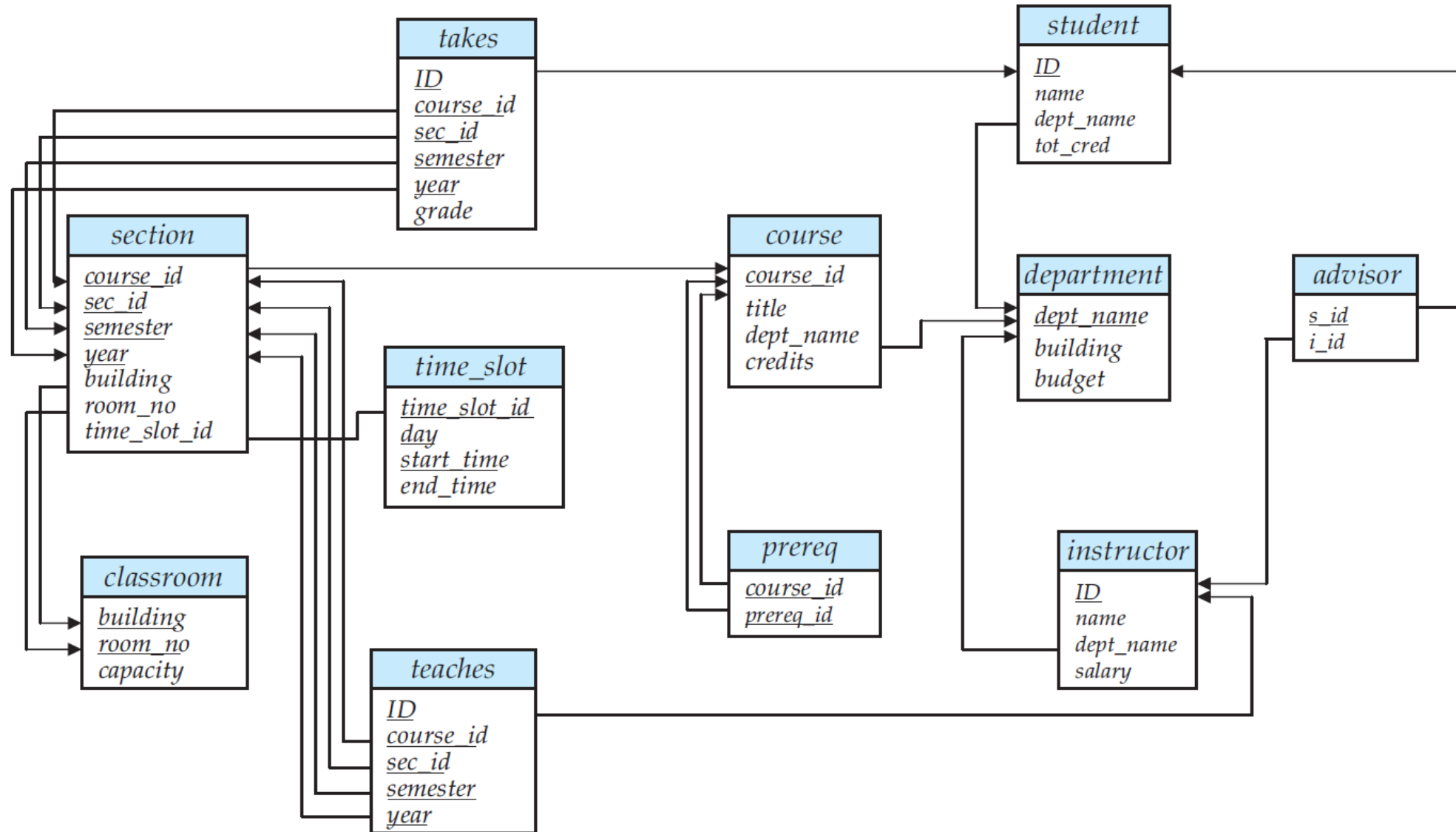
- Relational Algebra
- Tuple Relational Calculus
- Domain Relational Calculus

Relational Algebra

- Procedural language
- **Input:** one or two relations
- **Output:** a new relation
- **Operators:**
 - **Select:** σ
 - **Project:** Π
 - **Union:** $\cup$
 - **Intersection:** $\cap$
 - **Set difference:** $-$
 - **Cartesian product:** $\times$
 - **Rename:** ρ

Schema Diagram

University Database



Select Operation

- **Notation:**

$$\sigma_p(r)$$

- **Definition:**

$$\sigma_p(r) = \{t | t \in r \text{ and } p(t)\}$$

- p : a formula in propositional calculus consisting of **terms** connected by:

$$\wedge \text{ (**and**), } \vee \text{ (**or**), } \neg \text{ (**not**)}$$

- Each **term** is one of:

$$\langle \text{attribute} \rangle \text{ “}op\text{” } \langle \text{attribute} \rangle \text{ or } \langle \text{constant} \rangle$$

- “ Op ”: $=, \neq, >, \geq, <, \leq$
- Example: find all instructors in the Physics Department.

Project Operation

- **Notation:**

$$\Pi_{A_1, A_2, \dots, A_k}(r)$$

- A_1, A_2 : attribute names
- r : a relation
- **Output**: a relation of k columns obtained by erasing the columns that are not listed.
- **No duplicate** rows in output
- Example: To eliminate the *dept_name* attribute of *instructor*

Union Operation

- **Notation:**

$$r \cup s$$

- **Definition:**

$$r \cup s = \{t | t \in r \text{ or } t \in s\}$$

- **Conditions:**

1. r, s have the *same* **arity**
 2. **Compatible** attribute domains.
- Example: to find all courses taught in the Fall 2009 semester, or in the Spring 2010 semester, or in both

Set Difference Operation

- **Notation:**

$$r - s$$

- **Definition:**

$$r - s = \{t | t \in r \text{ and } t \notin s\}$$

- **Conditions:**

- r and s must have the **same arity**
 - attribute domains of r and s must be compatible
- **Example:** to find all courses taught in the Fall 2009 semester, but not in the Spring 2010 semester

Set-Intersection Operation

- **Notation:**

$$r \cap s$$

- **Definition:**

$$r \cap s = \{t | t \in r \text{ and } t \in s\}$$

- Conditions:
 - r, s have the **same arity**
 - Attributes of r and s are **compatible**
- Note: $r \cap s = r - (r - s)$

Cartesian-Product Operation

- **Notation**

$$r \times s$$

- **Definition:**

$$r \times s = \{tq | t \in r \text{ and } q \in s\}$$

- **Conditions:**

- Attributes of $r(R)$ and $s(S)$ are **disjoint**, e.g., $R \cap S = \phi$.
 - Rename if not disjoint.

Rename Operation

- **Notation:**

$$\rho_x(E)$$

- **Output:** the expression E under the name x

- In the case of arity:

$$\rho_{x(A_1, A_2, \dots, A_n)}(E)$$

- **Output:** the result of expression E under the name X , with the attributes renamed to $A_1, A_2, \dots, A_n$.

Formal Definition

- **Relational-algebra expressions:**

- $E_1 \cup E_2$

- $E_1 - E_2$

- $E_1 \times E_2$

- $\sigma_p(E_1)$

- $\Pi_s(E_1)$

- $\rho_x(E_1)$

Tuple Relational Calculus

- A **nonprocedural query language**,
- Query form:

$$\{t \mid P(t)\}$$

- The set of all tuples t , such that predicate P is true for t .
- t : **tuple variable**.
- $t \in r$: tuple t is in relation r .
- P : a **formula** similar to that of the predicate calculus.

Predicate Calculus Formula

1. Set of **attributes** and **constants**
2. Set of **comparison operators**: (e.g., $<$, $\leq$, $=$, $\neq$, $>$, $\geq$)
3. Set of **connectives**: and ($\wedge$), or ($\vee$), not ($\neg$)
4. **Implication** ($\Rightarrow$): $x \Rightarrow y$, if x is true, then y is true

$$x \Rightarrow y \equiv \neg x \vee y$$

5. Set of **quantifiers**:

- $\exists t \in r(Q(t))$
- $\forall t \in r(Q(t))$

Example

- Find the *ID*, *name*, *dept_name*, *salary* for instructors whose salary is greater than \$80,000.

Example

- Find the names of all instructors whose department is in the Watson building?

Example

- Find the set of all courses taught in the Fall 2009 semester, or in the Spring 2010 semester, or both?

Example Queries

- Find the set of all courses taught in the Fall 2009 semester, and in the Spring 2010 semester?

Example Queries

- Find the set of all courses taught in the Fall 2009 semester, but not in the Spring 2010 semester?

Universal Quantification

- Find all students who have taken all courses offered in the Biology department?

Safety of Expressions

- **Tuple calculus expressions** generating **infinite relations**.

- Example:

$$\{ t \mid \neg (t \in \textit{instructor}) \}$$

- **Output:** there are infinitely many tuples that are not in *instructor*.

- **Solution:** restriction on the set of allowable expressions.

$$\textit{dom}(t \in \textit{instructor} \wedge t[\textit{salary}] > 80000)$$

Safety of Expressions

- **Safe Expression:**

- Consider $\{t \mid P(t)\}$
- Every component of t appears in one of the relations, tuples, or constants that appear in P

- Example:

$$\{t \mid t[A] = 5 \vee true\} \quad \text{not safe}$$

Safety of Expressions

- Consider the query to find all students who have taken all courses offered in the Biology department:

Domain Relational Calculus

- A **nonprocedural query language** equivalent in power to the tuple relational calculus
- Query format:

$$\{ \langle x_1, x_2, \dots, x_n \rangle \mid P(x_1, x_2, \dots, x_n) \}$$

- $x_1, x_2, \dots, x_n$: domain variables
- P : a formula composed of atoms, similar to the predicate calculus

$$\exists x (P_1(x)) \text{ and } \forall x (P_1(x))$$

Example

- Find the *ID*, *name*, *dept_name*, *salary* for instructors whose salary is greater than \$80,000?

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Safety of Expressions

$$\{ \langle x_1, x_2, \dots, x_n \rangle \mid P(x_1, x_2, \dots, x_n) \}$$

- **Safety Conditions:**

1. All values that appear in tuples of the expression are values from ***dom*** (*P*)
2. For every “**there exists**” subformula of the form $\exists x (P_1(x))$, the subformula is true *iff* there is a value of x in *dom* (P_1) such that $P_1(x)$ is true.
3. For every “**for all**” subformula of the form $\forall_x (P_1(x))$, the subformula is true *iff* $P_1(x)$ is true for all values x from ***dom*** (***P1***).

Universal Quantification

- Find all students who have taken all courses offered in the Biology department?

Exercise

- **Relational algebra**
- Find the instructors in Physics with a salary greater than \$90,000:

Exercise

- **Relational algebra**
- Find all departments whose name is the same as their building name,

Exercise

- **Relational algebra**
- Find the name of all instructors in the Physics department.

Exercise

- **Relational algebra**
- Find the names of all instructors in the Physics department together with the course id of all courses they taught.

Exercise

- **Relational algebra**
- Find the largest salary in the university

Exercise

- **Relational algebra**
- Find the names of all instructors together with the course id of all courses they taught.”

Exercise

- **Tuple Relational Calculus**
- Find all students who have taken all courses offered in the CSE department.

Exercise

- **Domain Relational Calculus**
- Find all students who have taken all courses offered in the CSE department.



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